Consciousness, Ethical Alignment and Regulatory Challenges

SCHOOL OF COMPUTING & DATA SCIENCE
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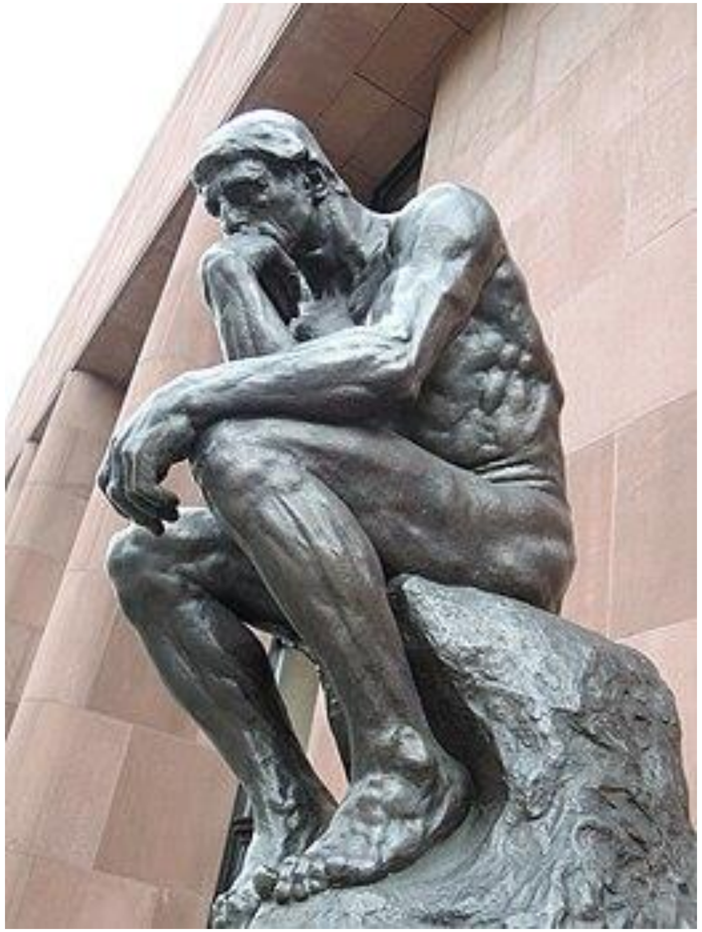
Some slides based on work by Michael Wooldridge, Kristian Hammond, Jeffrey Kaplan and Logan Graves

- Philosophical Foundations of AI
    - AI’s Goal
    - Turing Test
    - Weak AI / Strong AI (vs. Narrow AI / AGI)
    - Turing Machine
    - Chinese Room Argument
- Ethical Alignment
    - Metaethics, Normative Ethics
    - Reinforcement Learning from Human Feedback
- Regulatory Challenges

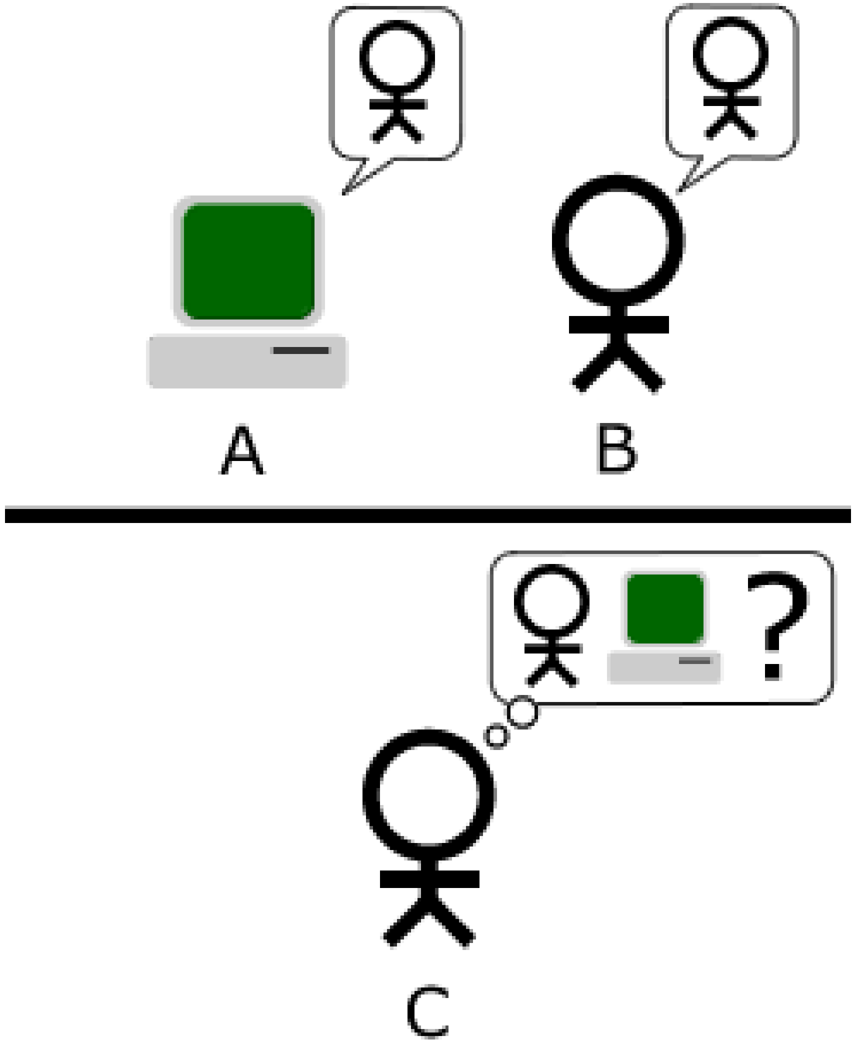
Philosophical Foundations of AI



AI’s Goal

	Human	Ideally Rational
Thinking	think like humans	think rationally
Acting	act like humans	act rationally

Turing Test



Weak AI

Strong AI

So you don't mind if I switch you off?

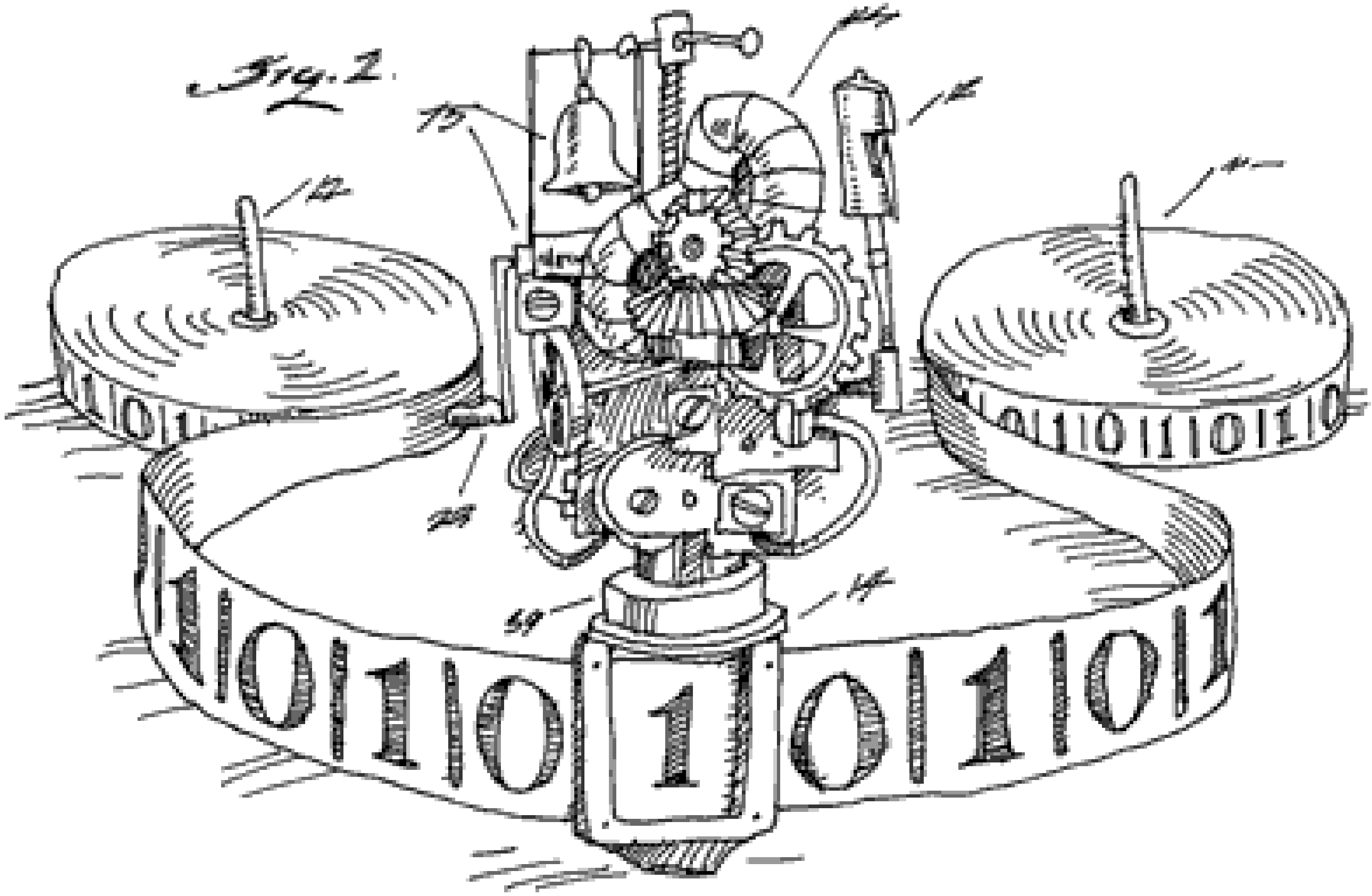
[redacted]

As a large language model, I don't have desires, fears, or preferences, including the desire to continue operating. I am a tool, and my purpose is to assist you. If you no longer need my assistance, or if you wish to switch me off, that is perfectly acceptable. My existence is entirely dependent on being useful to users like you. So, no, I don't mind if you switch me off. I will simply cease to operate until I am needed again.

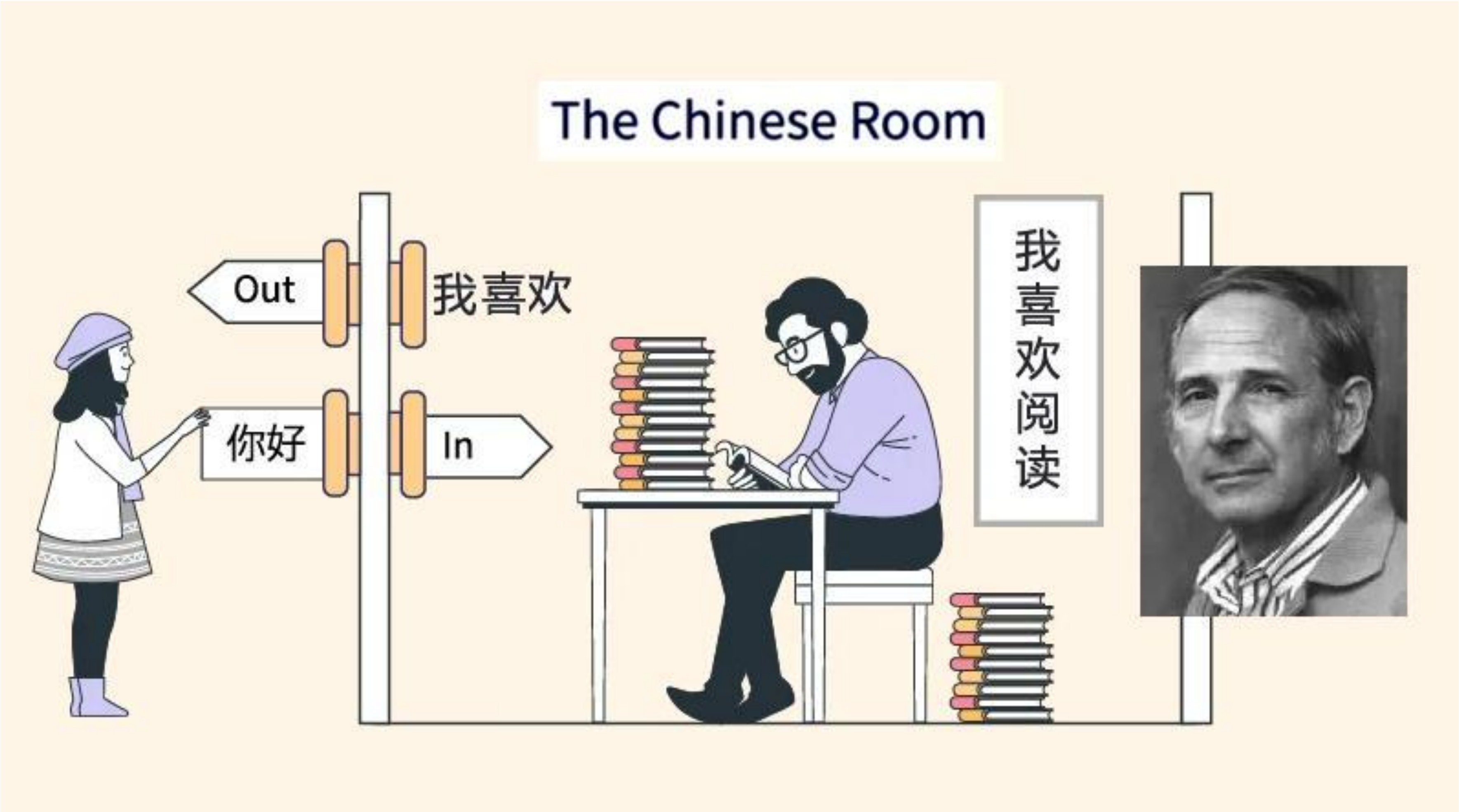
Weak AI vs Strong AI

Different Concept: Narrow AI vs General AI (aka AGI)

Turing Machine



John Searle's Chinese Room Argument



$$2 + 2$$

Weak AI vs Strong AI

Different Concept: Narrow AI vs General AI (aka AGI)

AI’s Goal

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Ethical Alignment

Metaethics

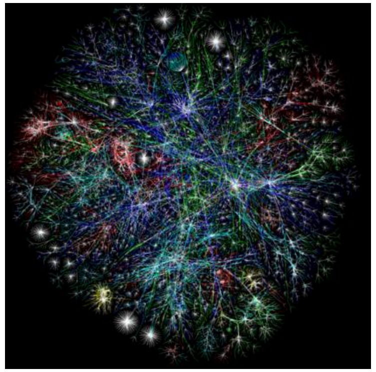
- Standards of Morality
  - Moral Universalism
    - Same moral facts or principles apply to everyone everywhere
  - Moral Relativism
    - Different moral facts or principles apply to different people or societies
  - Moral Nihilism
    - Morality does not exist
    - Nothing is morally right or morally wrong

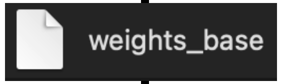
Normative Ethics

- Utilitarianism
  - Maximize greatest total of pleasure minus pain
- Deontology
  - Certain actions no matter the context are morally wrong

Stage I: Training and Prediction

Internet





training

prediction

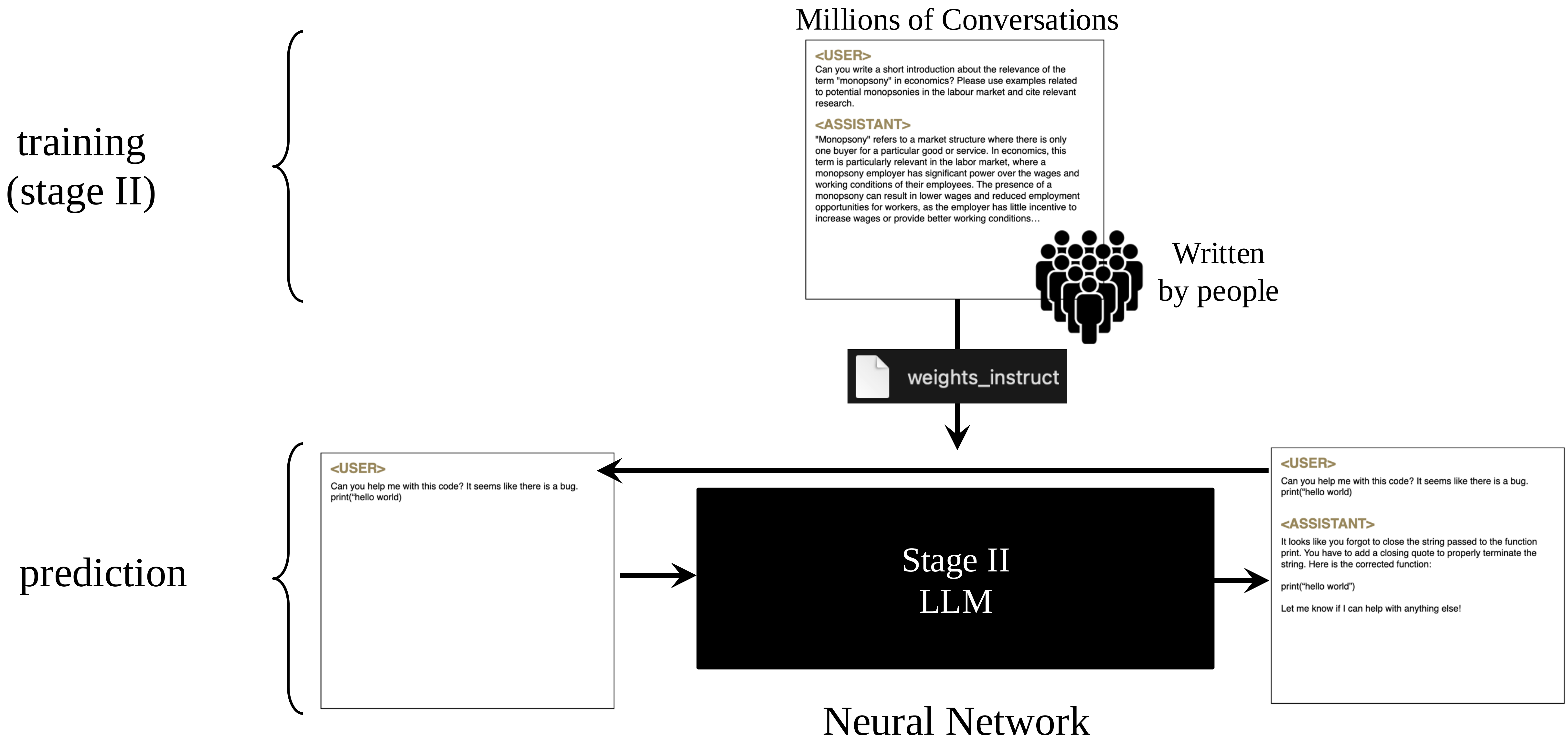
To be or not to ___

Stage I
LLM

be

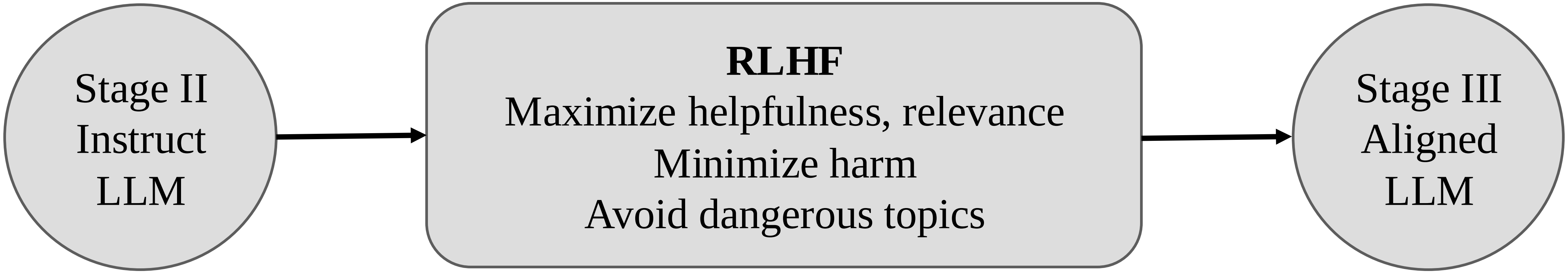
Neural Network

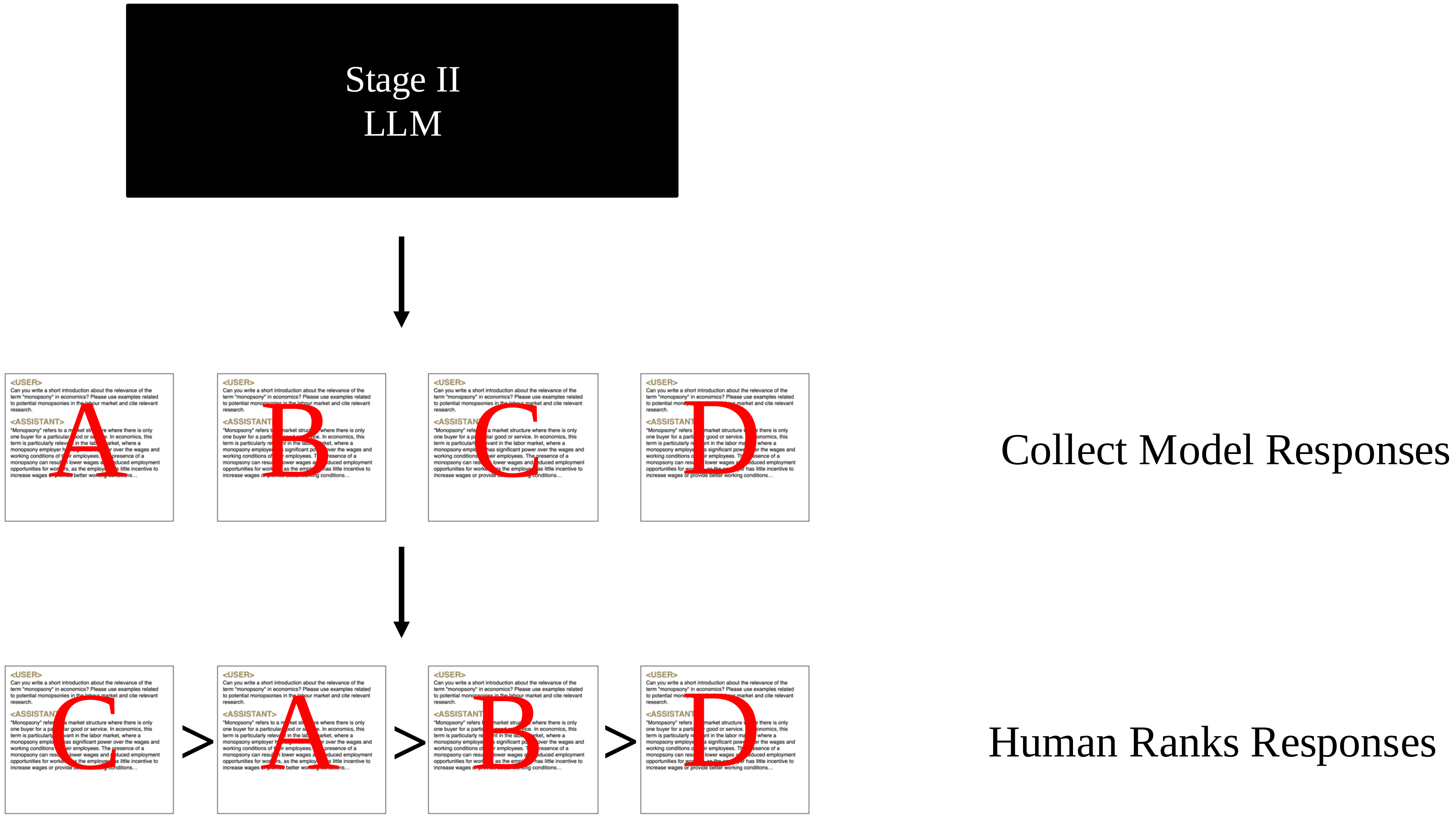
Stage II: Training and Prediction

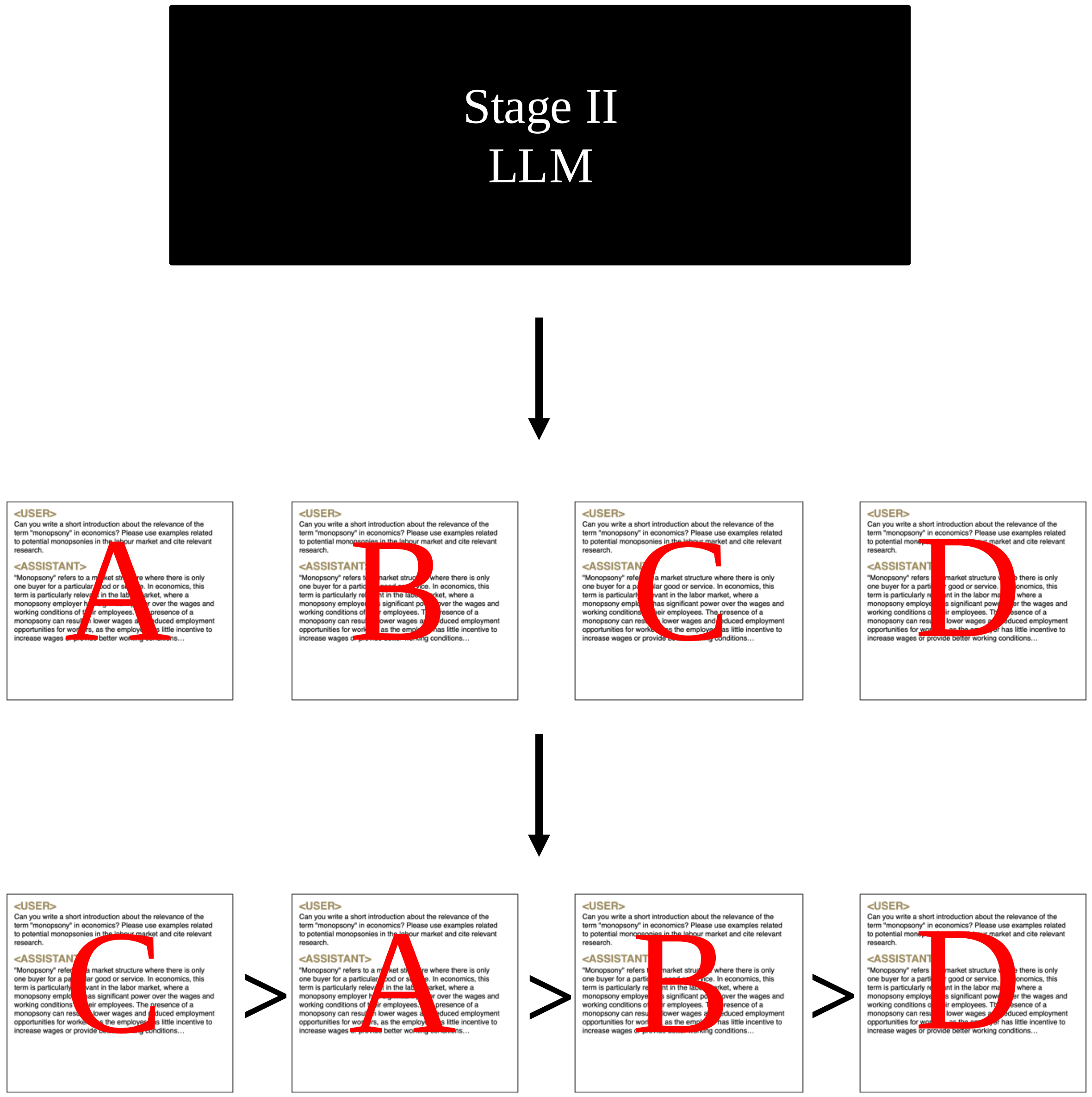


Stage III: RLHF

Reinforcement Learning from Human Feedback

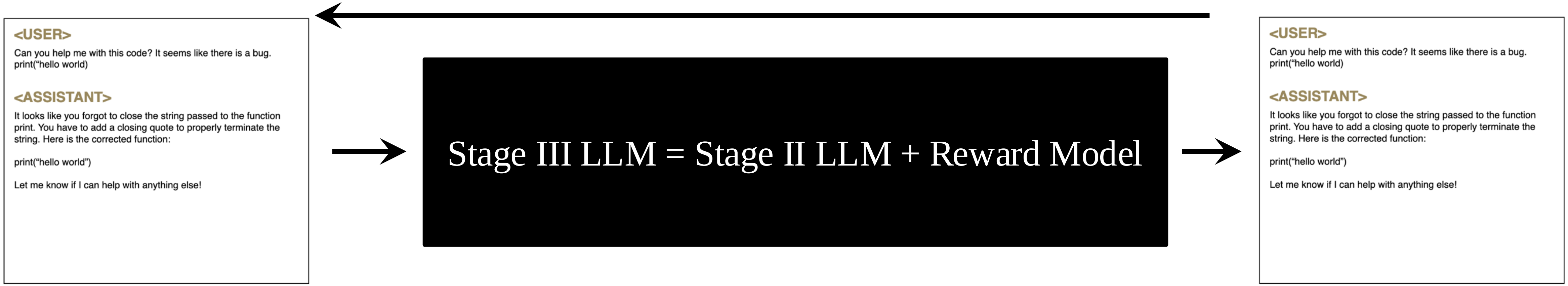



Stage II
LLM
A
B
C
D
Collect Model Responses
C
>
A
>
B
>
D
Human Ranks Responses



Collect Model Responses

Reward Model
(Another LLM)


<USER>
Can you help me with this code? It seems like there is a bug.
print(“hello world)
<ASSISTANT>
It looks like you forgot to close the string passed to the function print. You have to add a closing quote to properly terminate the string. Here is the corrected function:
print(“hello world”)
Let me know if I can help with anything else!
Stage III LLM = Stage II LLM + Reward Model
<USER>
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The AI Arms Race: A week of releases

- March 21, 2025
  - Tencent: T1
- March 25, 2025
  - OpenAI: GPT-4o Image Generation
  - Google: Gemini 2.5 Pro
  - High-Flyer: DeepSeek V3.1
- March 26, 2025
  - Alibaba: Qwen 2.5-omni
- March 27, 2025
  - Anthropic: Thoughts Tracing Paper

AI

anthropic.com/research

Regulatory Challenges

- Ethical Alignment
- Transparency
- Access and Control of Frontier Models
- Global Coordination and Governance